



Dangan (弾丸)

Progress documentation

A game by Jasper Steinberg

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1 PROJECT OVERVIEW



Dangan is a top-down bullet hell 2D shooter, inspired heavily by Touhou (having never played a Touhou game before, I think of *Dangan* as an introduction to the genre, to see if you like the gameplay). By getting high ranks on levels, you unlock more and progress through the game. Currently only 2 levels are playable.

The game mimics older hand-held games, since there are very few controls and you don't navigate with a mouse. You move with the arrow keys (or WASD), and shoot/select with SPACE. You cannot resize the game window; this is also by design.

As of writing this, there are two playable levels, but everything is fully functional. All designs are original and unique to *Dangan*. *Dangan* translates from Japanese to bullet, shot, or shell.

2 INITIAL PLAN + SKETCHES

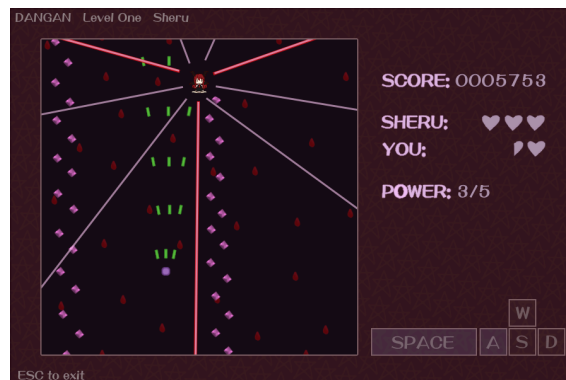
Starting off, I wanted to make a completely different game, namely a simple doodle-jump inspired game, where you jump from platform to platform staying above the lava underneath.



I scrapped this idea, since gravity was really tricky to calculate using deltatime, and so I scrapped the idea for *Dangan* (which has no gravity, and no collisions).

2.1 CONCEPT ART

Since I fell in love with the Touhou games' 'visual identity', I knew exactly how I wanted my game to look. I created these mockups when I only had the basic character movement, but stuck with the designs since I felt they were really pleasant to look at. I am happy that I decided to give each menu screen its own feel.



(nothing on these pages is interactive, everything is drawn on)

I also got a friend of mine to help me make the character art! I used their drafts in the mockups above.

While working on the project, whenever I needed a new art asset (such as a bullet or background), I would make it on the spot and implement it. I knew what art assets I needed, but decided to make them as I coded whenever I needed them so that I could focus on one thing at a time.

To help optimize the game, the background images are the exact resolution of the game screen (900x600).

2.2 CHARACTER ART

To guide my friend with the characters, I drew the following:



They took these and turned them into the final character artworks below:



The third character (the eldest sister) was meant to be the third and hardest level, but we ran out of time to draw and code for the third level.

3 DEVELOPMENT PROCESS

On every commit, I made clear when AI was used, and when it wasn't. The first mockup of the game, with movement, shooting, and a main menu was made without AI, but after leaving on holiday and working on other projects, I felt I had lost all my skills and decided to work with AI onwards.

My initial roadplan was:

Scene manager → Base functionality → Enemy attacks and game loop → Win/lose conditions → Multiple levels → Lore, dialogue, and any other changes
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I stayed true to this plan, but mainly because every step was based on the previous. In the end, I wasn't able to have 3 levels which I originally planned, and unfortunately there is no lore yet.

Most of the work happened within a week, which let me focus all of my time on the game, which I think helped refine my workflow for the whole project. I always made sure to push, pull, and update the GitHub often (which is why there are so many commits), and never lost any progress that way. I also got a new laptop near the beginning and had to migrate files.

4 CHALLENGES

Because I was coding with help from AI, I didn't want to let it make any decisions. All code that is written is purposeful and was consciously added. This meant that I needed to invent the logic for the level editing system. Most challenges were solved where I spoke to AI using pseudocode and it translated that into working python code. Then I would go in and rewrite, upgrade and tweak the code until it worked the way I wanted.

Some other challenges I faced during development apart from the level logic include everything surrounding the spinning blade, coming up with a point system, and ~~finding~~ making my own sound effects.

Ironically, working with AI was challenging, since Claude imposes a harsh limit on free users and how many messages you are allowed to send before running out of credits.

5 KNOWN LIMITATIONS

The game is not in a fully releasable state, mainly because only 2 levels exist.

Things I would change/add include:

- Adding lore (something along the lines of: three (evil) royal sisters have inherited the throne, and you must defeat them all to return peace to the kingdom)
- Completing all 5 levels
- More bullet movement types
- More settings (individual sound toggles, toggleable menu music, custom keybinds?)

I try creating things that work. Everything that you are able to do works flawlessly (as far as I know + on my laptop). I don't like cutting corners, but that means I need to sacrifice gameplay.

6 REFERENCES + AI DISCLOSURE

As mentioned, I used AI to help code. Many things that I wanted required a level of skill I don't have, but working with AI helped me bring the game to fruition. All comments in the code are my own and written at the very end without AI. The GitHub commits show which parts of the code were created with help from AI, namely Anthropic's Claude and Google's Gemini.

Regarding the game, the Manual page contains the game credits, but I will relist those below, along with some other points:

- Music from this game is from *PangYa!*. *PangYa* **was** a free-to-play MMO casual golf simulation video game, designed by South Korean development company Ntreev Soft for Microsoft Windows. The music in *Dangan* belongs to Ntreev Soft (but the *PangYa* soundtrack does not get flagged for copyright anywhere, and *PangYa! Reborn*, a fanmade copy of the game still running uses the same soundtrack).
- The rank system repurposes a concept from the manga *Made in Abyss*, used by mystical creatures to determine value. I decided to use the depictions and names since they fit in nicely with the rest of *Dangan*.
- The characters are original designs by a friend, going by 'moejion' online.

- The background images are made from historical copyright-free chinese paintings found on unsplash.com. Most are taken from museum archive accounts.
- In the manual, it says 'a game by juroimoh'. Juroimoh is the name of a creature from Made in Abyss (coincidentally from the village where the value system is used), and my username is on GitHub, so I have added it as a mini watermark.
- Tutorials were used for this project. I cannot confirm this is all of them, but I have listed the ones that I have saved:
 - ◆ <https://www.youtube.com/watch?v=tJiKYMqJnYg> (Masks)
 - ◆ <https://www.youtube.com/watch?v=4TfZjhw0J-8> (Sprites Explained)
 - ◆ <https://www.youtube.com/watch?v=-CJB0ipnx8s> (Collision Detection)
 - ◆ https://www.youtube.com/watch?v=mHyGHhs_nQg Code Structuring)
 - ◆ <https://www.youtube.com/watch?v=r0ixaTQxsUI> (Scenes)
- The game uses two fonts:
 - ◆ **DFPPOPCorn-W12** | <https://www.webfontfree.com/en/package/108555> (c)
Copyright DynaComware Corp. 2002
 - ◆ **VCR OSD MONO** | <https://www.dafont.com/mrmanet.d5509> Property of
Riciary Leal

7 FINAL THOUGHTS

I am very very very happy with how the game turned out, and might continue to complete the coding for all 5 levels if I have the motivation.

I'm happy to have made a game that is actually fun to play with an engaging point system. I spent most of my time coding in the Digital Media Lab, and my friends had a lot of fun playing the game while I was coding.

I enjoyed the Tech Basics I course a lot, especially the weekly challenges. Coding is intimidating, and it still is, but I feel a lot more capable in writing code and especially understanding it.